

Thermodynamics

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1 Energy and Enthalpy

1.1 Potential and Kinetic Energy

Core Concept 1.1: Thermodynamics

Thermodynamics examines the **energy changes** that accompany chemical reactions and phase transitions, enabling us to predict both whether a reaction will proceed **spontaneously** and **how much energy it will release or absorb**.

In chemistry, energy is broadly classified into two fundamental categories.

Core Concept 1.2: Kinetic Energy

Kinetic Energy (KE) refers to the energy associated with the motion of a molecule and is broadly categorized into **vibrational**, **rotational**, and **translational** motion.

Core Concept 1.3: Potential Energy

Potential Energy (PE) refers to the energy stored within intra- and intermolecular interactions. These interactions are governed by electrostatic attractions between opposing charges (e^- and protons) within a system, a principle formally described by **Coulomb's Law**.

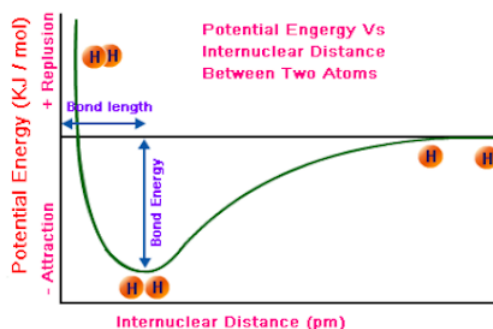


Figure 1: Bond potential energy diagram

As shown in Figure 1, intramolecular bonds arrange themselves at internuclear distances that **minimize PE**. When the internuclear distance is **infinitely large**, PE approaches zero, as the force between atoms is negligible.

As atoms approach one another, they reach an equilibrium point at which repulsive (*like-charge*) and attractive (*opposing-charge*) forces balance. The PE at this equilibrium is defined as the **bond energy**, and its formation is accompanied by a release of energy—analogous to two magnets snapping together. Conversely, compressing atoms beyond this equilibrium causes PE to rise, as repulsive forces begin to dominate—analogous to two rubber balls being forced together. *Bond energy is more rigorously characterized by the thermodynamic quantity **enthalpy**, discussed further in Core Concept 1.8.*

These observations give rise to two fundamental principles.

- **Breaking** a bond requires an **input of energy**; the change in PE from the bonded to unbonded state is therefore negative.
- **Forming** a bond releases an **output of energy**; the change in PE from the unbonded to bonded state is therefore positive.

Core Concept 1.4: The First Law of Thermodynamics

Within a **closed system**, the total energy is defined as the sum of its kinetic and potential energies:

$$E_{total} = KE + PE$$

This relationship reflects the **First Law of Thermodynamics**, which states that energy is neither created nor destroyed, but interconverted between KE and PE. When **PE decreases**—for instance, as two atoms form a bond and settle into a lower-energy configuration—the system gains a corresponding amount of KE. This conversion underlies **exothermic** reactions (Core Concept 1.5), in which the stabilization of chemical bonds manifests as increased molecular motion and the release of heat.

Conversely, when **KE decreases**—as when a molecule decelerates against an attractive intermolecular force—that energy is not lost, but transferred into PE as the molecule ascends a potential energy well shown in Figure 1

1.2 Heat

Core Concept 1.5: Endothermicity & Exothermicity

Chemical reactions can be classified based on the direction of heat flow between the system and its surroundings. In an **exothermic reaction**, energy is **released** into the surroundings—the products are at a lower energy state than the reactants, and the excess energy is dissipated as thermal energy. *Take the combustion of gasoline as an example.*

Conversely, in an **endothermic reaction**, energy is **absorbed** from the surroundings in order to drive the reaction forward—the products are at a higher energy state than the reactants, necessitating a net input of thermal energy. *A familiar example is the dissolution of ammonium nitrate in water; the principle underlying instant cold packs: as the reaction proceeds, it draws heat from the surrounding environment, producing a pronounced drop in temperature.*

Core Concept 1.6: Heat Transfer & Calorimetry

Heat transfer occurs when two systems at differing temperatures come into contact, whereupon thermal energy flows from the system of higher temperature to that of lower temperature.

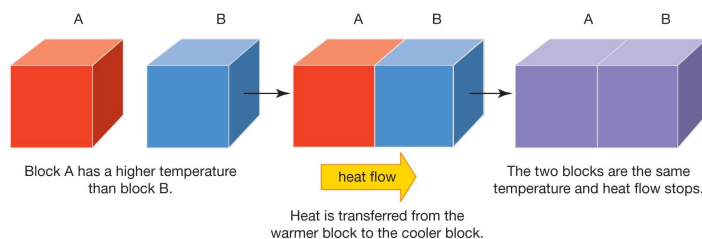


Figure 2: Heat transfer

This behavior reflects the underlying molecular basis of temperature: **hot** substances possess particles with greater average KE—vibrating and moving more rapidly—whereas **cold** substances possess particles with comparatively lower KE. **Temperature**, therefore, is more precisely defined as a measure of the **average kinetic energy** of the particles within a system.

The quantity of heat transferred within a system is given by:

$$q = m \times C \times \Delta T$$

where q denotes the heat transferred, m the mass of the system, C the **specific heat capacity**, and ΔT the change in temperature. **Specific heat capacity** is defined as the quantity of energy required to raise the temperature of 1 g of a substance by 1 K. For water, this value is $4.184 \text{ J g}^{-1} \text{ K}^{-1}$, meaning 4.184 joules or energy are required to raise the temperature of 1 g of water by 1 K.

The experimental discipline concerned with measuring the heat exchanged during chemical reactions, physical changes, or thermal interactions between objects is known as **calorimetry**, and constitutes a practical application of the **First Law of Thermodynamics**. Consider the following worked examples.

- (a) **Determine the heat required to raise the temperature of 43g of water by 45K.**

$$q = (43 \text{ g})(4.184 \text{ J g}^{-1} \text{ K}^{-1})(45 \text{ K}) = 8096.04 \text{ J}$$

While the joule is the standard unit of energy in chemistry, the **calorie** is another commonly encountered unit, defined as 4.184 J. Notably, this equivalence renders the specific heat capacity of water exactly $1 \text{ cal g}^{-1} \text{ K}^{-1}$. It should be noted that the dietary Calorie (Cal), as reported on nutritional labels, corresponds to 1 kilocalorie (kcal), or 1000 calories.

- (b) **A 560 g sample of Au at 100°C is placed into 560 mL of water at 20°C. Determine the final equilibrium temperature of the system. Given: $C_{\text{H}_2\text{O}} = 4.184 \text{ J g}^{-1} \text{ K}^{-1}$, $C_{\text{Au}} = 0.13 \text{ J g}^{-1} \text{ K}^{-1}$.**

By the conservation of energy, the total heat change within a closed system must equal zero:

$$q_{\text{lost}} + q_{\text{gained}} = 0 \implies q_{\text{gained}} = -q_{\text{lost}}$$

Since heat flows from the Au to the water, the Au loses energy while the water gains energy, and both reach a common final temperature T_{final} . Substituting into the calorimetry equation:

$$(560 \text{ g})(4.184 \text{ J g}^{-1} \text{ K}^{-1})(T_{\text{final}} - 20) = -(560 \text{ g})(0.13 \text{ J g}^{-1} \text{ K}^{-1})(T_{\text{final}} - 100)$$

Solving for T_{final} :

$$T_{\text{final}} = 22.41^\circ \text{C}$$

Core Concept 1.7: Phase Transitions

Phase transitions refer to the physical transformation of a substance from one state of matter to another, driven by changes in thermal energy. The heat required to transition water from a solid to a liquid state is termed the **heat of fusion**, while that required to transition water from a liquid to a gaseous state is termed the **heat of vaporization**.

The sign of each quantity is determined by the direction of the transition: positive for melting and vaporization (endothermic), and negative for freezing and condensation (exothermic).

As illustrated in Figure 3, temperature remains constant during phase transitions. During vaporization, for instance, the energy supplied does not increase the average KE of the molecules (*which temperature measures*),

but is instead directed toward increasing the **PE of the IMFs**. At 100°C, water molecules in the liquid phase possess the same average KE as those in the gas phase; however, they remain confined to the liquid state by the deep PE wells created by hydrogen bonding interactions. The heat of vaporization, therefore, represents **the energy required to overcome these IMFs entirely**, liberating the molecules from their neighbors and enabling the transition to the gas phase. *The same reasoning applies analogously to the heat of fusion.*

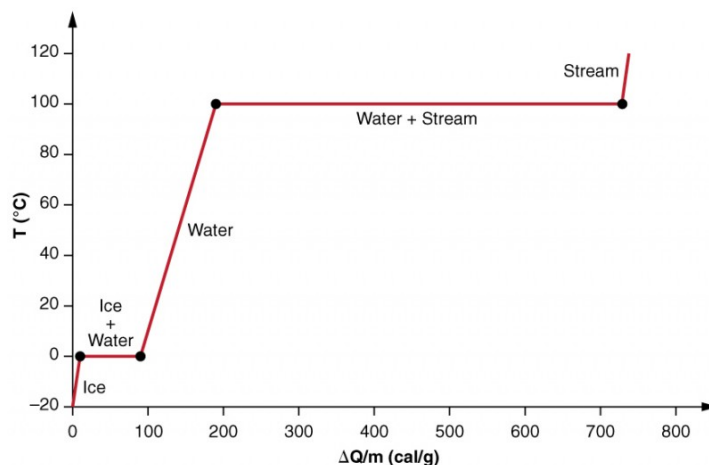


Figure 3: Heating curve of water

It is also notable that the heat of vaporization is substantially greater in magnitude than the heat of fusion. This disparity arises because **fusion** requires only sufficient energy to **disrupt the rigid, ordered crystal lattice** of ice—partially weakening intermolecular interactions without eliminating them—whereas **vaporization** necessitates the **complete dissociation** of all intermolecular interactions between water molecules. The energetic difference between these two processes is therefore considerable.

1.3 Enthalpy

Core Concept 1.8: Enthalpy

Enthalpy (H) is a thermodynamic state function representing the **total heat content** of a system at constant pressure. Formally, it is defined as the sum of the system's internal energy and the pressure-volume work required to establish its physical extent within the surrounding environment—that is, *the energy expended to displace the atmosphere and “make room” for the system.*

Although the absolute enthalpy of a system cannot be determined experimentally, the **change in enthalpy** (ΔH) during a chemical or physical process is directly measurable, making it one of the most practically significant quantities in thermochemistry.

To understand the molecular origin of ΔH , recall the potential energy diagram (Figure 1) introduced in Core Concept 1.3.

At the molecular level, all chemical reactions involve the breaking of bonds in the reactants and the formation of new bonds in the products. Since **bond formation releases energy** and **bond breaking requires energy**, the net enthalpy change of a reaction reflects the difference between these two contributions. When the energy released upon forming product bonds exceeds the energy required to break reactant bonds, the reaction is classified as **exothermic** ($\Delta H < 0$); conversely, when more energy is consumed than released, the reaction is **endothermic** ($\Delta H > 0$).

Core Concept 1.9: Enthalpy Example (Combustion of Hydrogen)

Take the combustion of hydrogen to produce water ($2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$) as an illustrative example. The enthalpy change over the course of this reaction is depicted below.

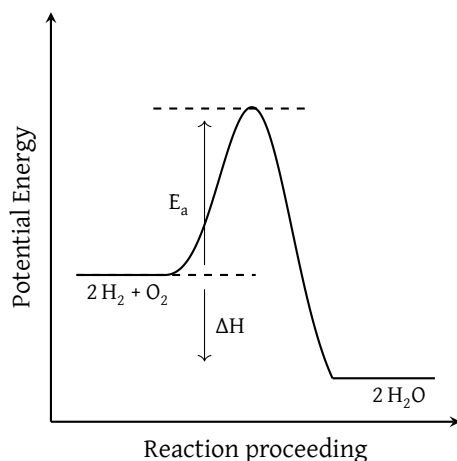


Figure 4: Enthalpy diagram of H_2 combustion

To build intuition, consider the reaction as it would appear in practice. Upon igniting a hydrogen-filled balloon, one observes an immediate release of heat and light—a clear indication that the reaction is exothermic. The original reactants (H_2 within the balloon and O_2 from the surrounding air) are converted into the product (H_2O) following an initial input of energy supplied by the ignition source. This initial energy input is termed the **activation energy** (E_a), which is the minimum energy required to **initiate the breakdown of reactant bonds** before product bonds can form.

The fundamental driving force underlying chemical reactivity is **the tendency of a system to minimise its potential energy**. Consequently, the free atoms produced upon bond dissociation—H, H, H, H, O, O—reorganise to form $2\text{H}_2\text{O}$, whose bonds exist at a lower potential energy state than those of the original reactants. The net difference in enthalpy between reactants and products is released as heat, manifesting as the observed explosion.

It is worth noting that a single spark does not supply sufficient activation energy to convert all H_2 and O_2 molecules simultaneously. Rather, the initial spark provides enough energy to initiate the formation of a single H_2O molecule. The energy released by this event then serves as the activation energy for neighbouring reactant molecules, which in turn release energy that drives further reactions. This cascade propagates rapidly throughout the balloon, accounting for the near-instantaneous nature of the explosion.

The enthalpy change of a reaction can also be determined quantitatively using average bond enthalpies. Representative values for several common bonds are given in Table 1.

Bond	H-H	O-O	H-O	C-C	C=C	$\text{C}\equiv\text{C}$	C-O	O=O
H_{average} (kJ/mol)	436	146	463	348	612	838	360	498

Table 1: Average bond enthalpies of common bonds

Applying these values to the combustion of hydrogen, the enthalpy change is calculated by summing the energies of bonds broken (*endothermic, positive sign*) and bonds formed (*exothermic, negative sign*):

- Break 2 mol of H-H $\rightarrow (2)(+436 \text{ kJ mol}^{-1})$

- Break 1 mol of O=O $\rightarrow$ (1)(+498 kJ mol⁻¹)
- Form 4 mol of O-H $\rightarrow$ (4)(-463 kJ mol⁻¹)

Thus $\Delta H = -482 \text{ kJ mol}^{-1}$ for the reaction. Finally, it is important to note that E_a does not represent the energy required to break all reactant bonds simultaneously. E_a is **only the energy needed to break key reactant bonds to the extent necessary for the reaction to proceed**. For instance, the activation energy for water formation is approximately 200 kJ mol⁻¹ – far less than the 1370 kJ mol⁻¹ that would be required for complete dissociation of all reactant bonds.

Core Concept 1.10: Enthalpy Example (Autoionization of Water)

Consider the autoionization of water as another example.

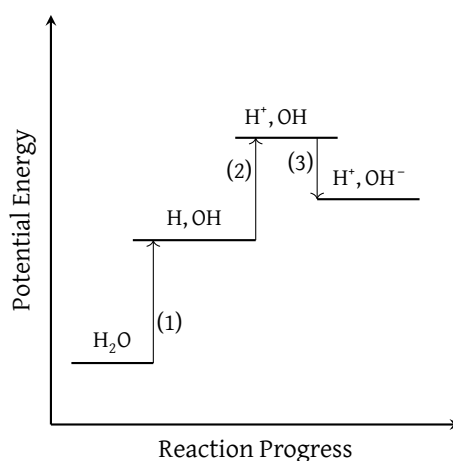


Figure 5: Enthalpy diagram for water autoionization

The diagram above in Figure 5 shows a net rise in PE, indicating that the overall reaction is endothermic ($\Delta H > 0$). The process can be decomposed into three distinct steps, each with an associated enthalpy contribution:

- (1) Break 1 mol of O-H $\rightarrow$ (1)(+467 kJ mol⁻¹)
- (2) Remove e⁻ from 1 mol of H (ionisation energy) $\rightarrow$ (1)(+1312 kJ mol⁻¹)
- (3) Add e⁻ to 1 mol of O (electron affinity) $\rightarrow$ (1)(-180 kJ mol⁻¹)

Thus $\Delta H = +1599 \text{ kJ mol}^{-1}$ for the reaction.

Core Concept 1.11: Hess's Law

When two or more chemical equations are combined to yield an overall reaction, the enthalpy of that reaction is equal to the sum of the enthalpies of the constituent steps. This principle is known as **Hess's Law**, and it follows directly from the fact that enthalpy is a **state function**—one whose value depends only on the initial and final states of the system, not on the path taken between them.

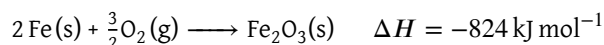
$$\Delta H_{\text{rxn}} = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$$

Conceptually, this expression corresponds to notionally **reversing the formation of the reactants**

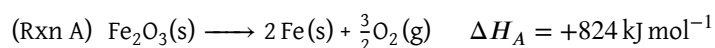
(contributing a positive enthalpy term) and then **carrying out the formation of the products** (contributing a negative enthalpy term), with the net enthalpy change reflecting the difference between the two.

Core Concept 1.12: Enthalpy Example (Iron Smelting)

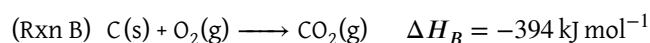
Consider the reaction of iron oxide with carbon. The formation of iron oxide is an exothermic process, as given by the following equation.



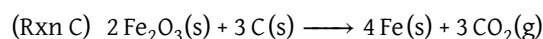
By Hess's Law, the reverse reaction—the decomposition of iron oxide—is endothermic with an equal and opposite enthalpy change. This is designated **Rxn A**.



Next, consider the combustion of carbon, designated **Rxn B**.



Combining 2 equivalents of Rxn A with 3 equivalents of Rxn B yields the overall reduction of iron oxide by carbon, designated **Rxn C**.



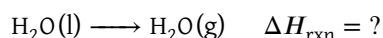
Applying Hess's Law, the enthalpy of Rxn C is obtained by summing the scaled enthalpies of Rxn A and Rxn B.

$$\Delta H_C = 2\Delta H_A + 3\Delta H_B = +466 \text{ kJ mol}^{-1}$$

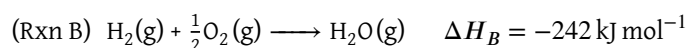
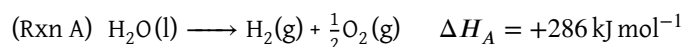
This result indicates that deoxidising 2 moles of iron oxide requires $+466 \text{ kJ mol}^{-1}$, or equivalently $+233 \text{ kJ mol}^{-1}$ per mole—a substantially more energetically favourable outcome than the $+824 \text{ kJ mol}^{-1}$ that would be required in the absence of carbon. This reaction forms the thermochemical basis of the iron smelting process, by which raw iron ore is reduced to pure iron.

Core Concept 1.13: Enthalpy Example (Vaporization of Water)

Consider the vaporization of water at 25°C , in which liquid water is converted to its gaseous phase.



This transformation can be decomposed into two intermediate steps. Rxn A represents the decomposition of liquid water into its constituent elements, and Rxn B represents the formation of gaseous water from those elements.



Applying Hess's Law, the enthalpy of vaporization is obtained by summing the enthalpies of the two steps.

$$\Delta H_{\text{rxn}} = \Delta H_A + \Delta H_B = +44 \text{ kJ mol}^{-1}$$

This result is consistent with the general formulation introduced previously, whereby the overall enthalpy change corresponds to **reversing the formation of the reactants** and **carrying out the formation of the products**.

$$\Delta H_{\text{rxn}} = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$$

2 Entropy

2.1 Defining Entropy

Enthalpy alone is insufficient to predict whether a reaction will proceed spontaneously. The dissociation of ammonium chloride into its constituent ions, for instance, is endothermic yet proceeds to completion.



Similarly, the spontaneous expansion of a gas into a vacuum occurs irreversibly despite $\Delta H = 0$, and thus cannot be accounted for by enthalpic considerations alone.

Core Concept 2.1: Entropy & The Second Law of Thermodynamics

There exists a natural tendency for energy and matter to disperse—in thermodynamic terms, systems tend to evolve towards states of higher **entropy**. This principle constitutes **the second law of thermodynamics**. Although introductory texts often characterise entropy as *disorder*, this description is imprecise. More rigorously, entropy is defined as a measure of the number of distinct microscopic configurations (**microstates**) consistent with a given observable state (**macrostate**).

Core Concept 2.2: Entropy (Dice Analogy)

Consider the sum obtained from rolling two dice. Across all 36 possible outcomes, the probability distribution is strongly concentrated around a sum of 7. In this context, the macrostate is defined by the sum, while the microstates are the individual dice combinations that yield it (6-1, 5-2, 4-3, 3-4, 2-5, 1-6). A low-entropy macrostate such as a sum of 2 has only a single accessible microstate, whereas the sum of 7 has six—making high-entropy macrostates statistically far more probable.

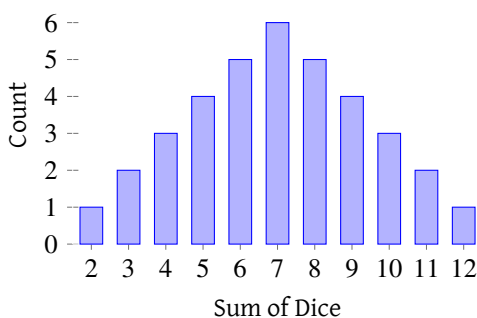


Figure 6: Dice roll probability distribution

Core Concept 2.3: Entropy (Gas Molecule Traits)

Consider a collection of gas molecules confined within a closed container. The spontaneous dispersion of the gas throughout the available volume is directly analogous to the dice example: uniform distribution across the container corresponds to the highest-entropy macrostate, as it is accessible through the greatest number of positional microstates. By contrast, the configuration in which all molecules are concentrated in a single corner of the container represents a low-entropy macrostate, accessible through comparatively few microstates. This is illustrated at a reduced scale in Figure 7.

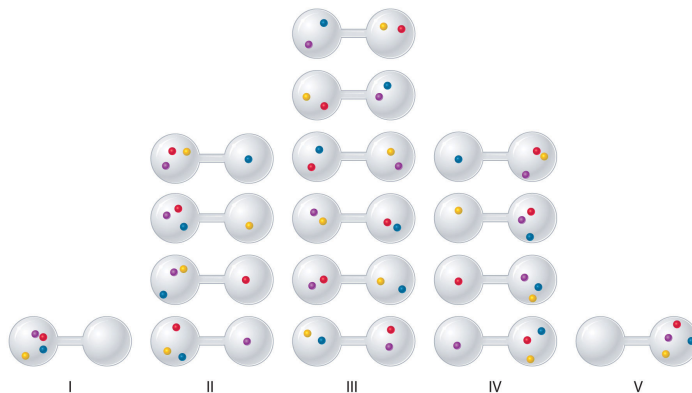


Figure 7: Gas molecule position microstates

The preceding discussion has focused exclusively on the positional coordinates of gas molecules. In reality, the number of accessible microstates for a given macrostate is further expanded by additional degrees of freedom, including rotational orientation, vibrational motion, translational velocity, and KE magnitude. Those dimensions associated with KE are, by extension, coupled to temperature. Just as positional microstates favour uniform spatial distribution, the microstates associated with KE favour its broad dispersal throughout the system—providing a statistical basis for the spontaneous transfer of heat from warmer to cooler regions.

It is important to note that when gas molecules are described as being at 25 °C, this refers to the *average* KE of the ensemble—not to all molecules simultaneously possessing the KE equivalent of 25 °C. At any given instant, individual molecules will deviate from this average, with some possessing slightly higher and others slightly lower KEs. Consequently, there exist far more microstates in which KEs are broadly distributed around the mean than those in which all molecules possess identical KEs.

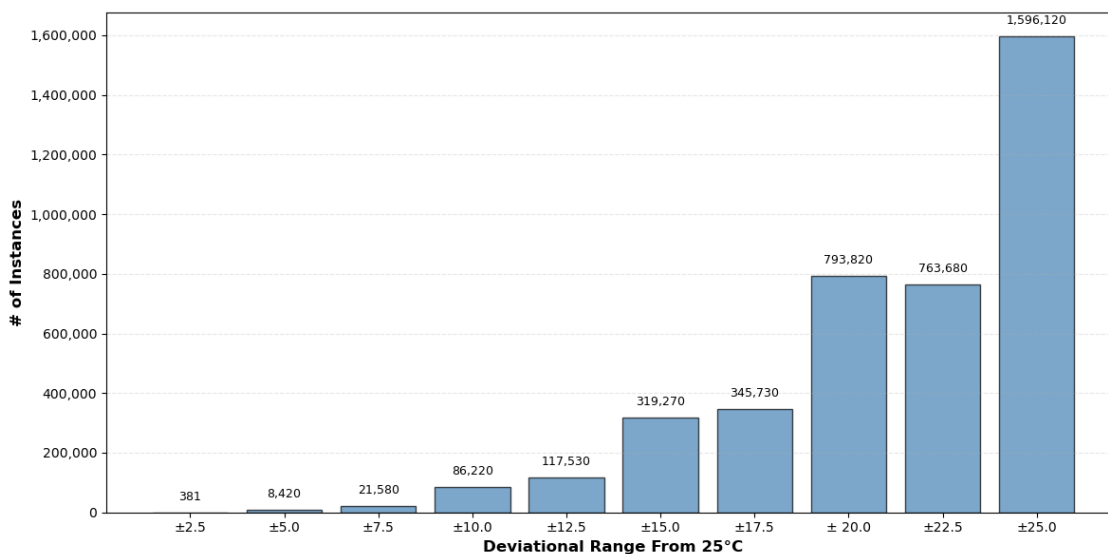


Figure 8: Gas kinetic energy distribution

Figure 8 plots the number of possible 5-integer permutations that sum to 125 (*and thus average to 25*), partitioned by the maximum permitted deviation from the mean. Even in this highly simplified model, the number of accessible microstates increases dramatically as the permitted spread in KE widens—vastly outnumbering the

single microstate in which all values equal 25, such as the permutation (25, 25, 25, 25, 25). This statistical tendency for KE to be broadly and randomly distributed among gas molecules, while the ensemble average remains fixed, is the basis of the **Maxwell–Boltzmann Distribution** (Core Concept 1.2).

Core Concept 2.4: Boltzmann Equation

Entropy (S) is formally quantified through the **Boltzmann equation**:

$$S = k_B \ln(W)$$

where W denotes the number of energetically equivalent microstates accessible to the system, and k_B is the Boltzmann constant, defined as

$$k_B = \frac{R}{N_A} = \frac{8.314 \text{ J K}^{-1} \text{ mol}^{-1}}{6.022 \times 10^{23} \text{ mol}^{-1}} = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

where R is the ideal gas constant and N_A is Avogadro's number. Physically, k_B represents the amount of thermal energy contained per particle per degree of temperature.

As the preceding discussion illustrates, the number of accessible microstates grows exponentially with the number of particles in the system. It is for this reason that entropy is proportional to $\ln(W)$ rather than W directly—a relationship revisited in Core Concept 3.5.

In more general terms, entropy increases ($\Delta S > 0$) when

- (a) Temperature increases.
- (b) A reaction produces a net increase in the number of molecules.
- (c) A phase transition occurs toward a less ordered phase (solid $\rightarrow$ liquid or liquid $\rightarrow$ gas).

In more nuanced cases, the guiding principle remains whether the process increases or decreases the number of accessible microstates.

2.2 Changes in Entropy

Core Concept 2.5: The Third Law of Thermodynamics

The **third law of thermodynamics** states that as the temperature of a system approaches absolute zero (0 K), its entropy approaches a constant minimum value—defined as zero for a perfect, pure crystal. This is significant because it provides an absolute reference point from which the entropy of any system, molecule, or quantity of substance can be determined. This stands in contrast to enthalpy, for which no such absolute reference exists and only changes can be meaningfully quantified.

Core Concept 2.6: Entropy & Temperature

Entropy is expressed in units of **Joules per Kelvin (J K^{-1})**. This unit is best understood by analogy: just as density (g L^{-1}) describes how much mass a substance occupies per unit volume, entropy describes the breadth of accessible microstates—quantified in units of energy—per unit temperature.

It is also important to note that **the entropy of a system increases with temperature, but at a diminishing rate**. To build intuition, consider supplying an equal quantity of energy to boiling water and to ice. Since

boiling water already exists in a high-entropy state, a 1 K rise in temperature produces a proportionally smaller increase in entropy compared to the same temperature rise applied to ice, which begins in a low-entropy state. This relationship is expressed mathematically as

$$\frac{dS}{dT} = \frac{C_p}{T}$$

This expression states that the rate of entropy change with respect to temperature is proportional to the heat capacity of the system (C_p) and inversely proportional to the temperature (T). The inverse dependence on T reflects the diminishing effect described above. The proportionality to C_p follows from the fact that a higher heat capacity requires a greater energy input per kelvin of temperature increase—and consequently generates a greater expansion of accessible microstates. For instance, raising the temperature of water by 1 K requires substantially more energy than the same increase in copper; this large energy input into water produces a correspondingly greater increase in molecular disorder than the comparatively modest input required for copper.

Core Concept 2.7: Thought Experiment (*Increasing the Temperature of Water*)

Consider the heating of water from 25 °C to 50 °C, examined through the dual lenses of enthalpy and entropy.

From an enthalpic standpoint, this is a straightforward calorimetric calculation: ΔH is given by the product of the molar quantity of H_2O , its heat capacity C_{H_2O} , and the temperature change ΔT .

To express the same energy input in terms of entropy, two factors must be accounted for simultaneously: (1) the monotonic increase in entropy with temperature, and (2) the diminishing rate at which this increase occurs. These conditions are summarised mathematically as

$$\frac{dS}{dT} > 0, \quad \frac{d^2S}{dT^2} < 0$$

The entropy gained over this temperature interval corresponds to the area under the S versus T curve between the initial and final temperatures, reflecting the cumulative expansion of accessible microstates. The functional form of this curve resembles $f(x) = \ln(x)$, consistent with the constraints above.

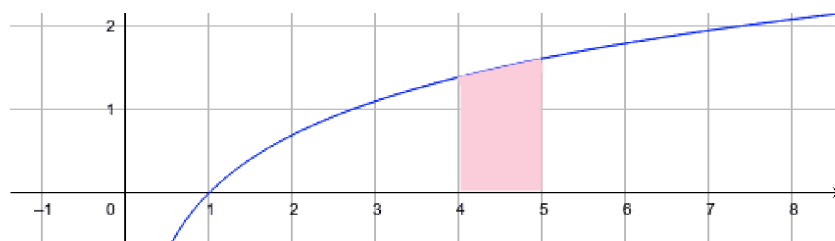


Figure 9: Entropy vs temperature

On closer inspection, enthalpy can be understood as a macroscopic surrogate for the entropy change of the *surroundings*—that is, the thermal, temperature-dependent component of total entropy change. A complete thermodynamic description requires additionally accounting for the entropy change of the *system* itself, which captures configurational and positional contributions that enthalpy is blind to. It is the sum of these two contributions that ultimately determines spontaneity—a relationship formalized by Gibbs free energy (Core Concept 3.2).

3 Gibbs Free Energy

3.1 Gibbs Free Energy Foundations

Core Concept 3.1: Deriving the Gibbs Free Energy Equation

Returning to the **second law of thermodynamics**—which states that the total entropy of an isolated system increases over time—and treating the universe as an isolated system, this law can be expressed as the following.

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{environment} > 0$$

To illustrate, consider the melting of 10 moles of ice. Since temperature remains constant during a phase transition, the entropy change of the system can be calculated directly from the enthalpy of fusion without integration.

$$\Delta S_{system} = \frac{\Delta H_{fusion}}{T_{system}} = \frac{(6.02 \text{ kJ mol}^{-1})(10 \text{ mol})}{273 \text{ K}} = +220 \text{ J K}^{-1}$$

The temperature of the surroundings does technically decrease upon absorbing heat from the system, but this change is negligible given that the surroundings are modelled as a **thermal reservoir**—a body so large that its heat capacity is effectively infinite relative to the system, and its temperature remains constant at 298 K.

The corresponding entropy change of the surroundings is therefore

$$\Delta S_{surroundings} = \frac{-\Delta H_{fusion}}{T_{surroundings}} = \frac{-(6.02 \text{ kJ mol}^{-1})(10 \text{ mol})}{298 \text{ K}} = -202 \text{ J K}^{-1}$$

The total entropy change of the universe is then

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} = 220 \text{ J K}^{-1} + (-202 \text{ J K}^{-1}) = +18 \text{ J K}^{-1}$$

This result reflects two competing contributions.

- (1) The entropy of the system increases upon absorption of heat.
- (2) The entropy of the surroundings decreases as heat is transferred from it—**but to a lesser degree**, since the same quantity of energy represents a smaller change at the higher temperature of the surroundings.

Algebraically manipulating the expression for $\Delta S_{universe}$ yields the **Gibbs free energy** equation. Substituting

$$\Delta S_{surroundings} = -\frac{\Delta H_{sys}}{T_{surr}}:$$

$$\Delta S_{uni} = \Delta S_{surr} + \Delta S_{sys} > 0$$

$$\Delta S_{uni} = \frac{-\Delta H_{sys}}{T_{surr}} + \Delta S_{sys} > 0$$

$$T_{surr} \Delta S_{uni} = -\Delta H_{sys} + T_{surr} \Delta S_{sys} > 0$$

$$-T_{surr} \Delta S_{uni} = \Delta H_{sys} - T_{surr} \Delta S_{sys} < 0$$

Defining $\Delta G = -T_{surr} \Delta S_{uni}$, the condition for spontaneity becomes

$$\Delta G = \Delta H_{sys} - T_{surr} \Delta S_{sys} < 0$$

Core Concept 3.2: Gibbs Free Energy

Gibbs free energy (ΔG) provides a unified criterion for thermodynamic spontaneity, effectively quantifying the net tendency of the universe's entropy to increase. It represents a balance between two competing thermodynamic driving forces.

- (1) The tendency to maximise the entropy of the universe
- (2) The tendency to minimise the enthalpy of the system.

When both conditions are satisfied simultaneously, the reaction is unconditionally **spontaneous**; when neither is satisfied, the reaction is unconditionally **non-spontaneous**; when only one is satisfied, spontaneity is determined by the relative magnitudes of the two contributions.

ΔH°	ΔS°	T	ΔG°	Reaction Favorability
–	+	ANY	–	Product favored ($\rightarrow$)
–	–	HIGH	+	Reactant favored ($\leftarrow$)
–	–	LOW	–	Product favored ($\rightarrow$)
+	+	HIGH	–	Product favored ($\rightarrow$)
+	+	LOW	+	Reactant favored ($\leftarrow$)
+	–	ANY	+	Reactant favored ($\leftarrow$)

Table 2: Reaction favorability across different ΔH and ΔS values

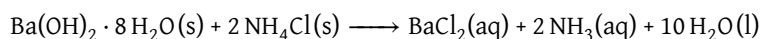
ΔG_{rxn} quantifies the difference in free energy between products and reactants under standard-state conditions, and—like enthalpy and entropy—is a state function. It may be calculated either from the corresponding ΔH_{rxn} and ΔS_{rxn} values, or directly from tabulated standard Gibbs free energies of formation (ΔG_f).

$$\Delta G_{rxn} = \sum \Delta G_f(\text{products}) - \sum \Delta G_f(\text{reactants})$$

A useful way to interpret the Gibbs free energy equation is to treat $T\Delta S$ as a temperature-dependent scaling term that modulates the entropic contribution to spontaneity—amplifying or diminishing the influence of entropy depending on the thermal conditions of the system.

Core Concept 3.3: Gibbs Free Energy Example (Barium Hydroxide & Ammonium Chloride Reaction)

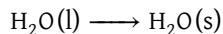
Consider the reaction between barium hydroxide octahydrate and ammonium chloride.



Although the reaction is highly endothermic ($\Delta H = +62 \text{ kJ mol}^{-1}$), inspection of the physical states of the reactants and products reveals a substantial increase in entropy ($\Delta S = +368 \text{ J mol}^{-1} \text{ K}^{-1}$), driven by the increase in the number of moles of molecules and the transition to more disordered phases. At 298 K, the magnitude of the entropic contribution $T\Delta S = +109.7 \text{ kJ mol}^{-1}$ exceeds the enthalpic penalty, yielding $\Delta G < 0$ and confirming that the reaction proceeds spontaneously.

Core Concept 3.4: Thought Experiment (*Liquid Water at Absolute Zero*)

Consider the behaviour of water at a theoretical temperature of 0 K. It is well established that liquid water freezes at absolute zero, but the thermodynamic basis for this warrants examination. The relevant phase transition is represented by the following equation.



Freezing is exothermic and therefore enthalpically favourable. Although the transition from liquid to solid entails a decrease in entropy, this entropic penalty carries no thermodynamic weight at absolute zero—since $T\Delta S = 0$ when $T = 0 \text{ K}$, the entropic contribution to ΔG vanishes entirely. At absolute zero, Gibbs free energy reduces to enthalpy alone, and the exothermic nature of freezing makes the solid state unconditionally favoured. Note that while liquid water cannot physically exist at 0 K, this theoretical limiting case is precisely what the thermodynamic framework predicts and seeks to explain.

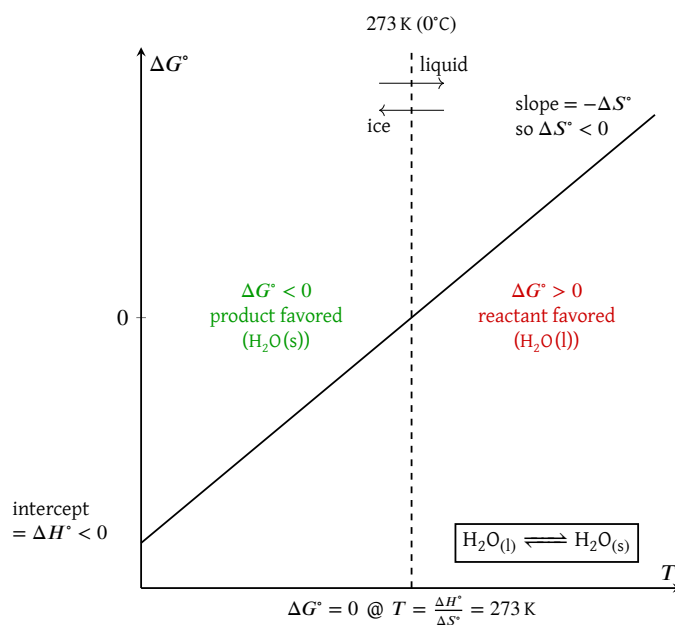


Figure 10: Gibbs Free Energy of Water

As temperature increases, the $T\Delta S$ term grows in magnitude, progressively amplifying the entropic penalty associated with freezing. Below 273 K, this penalty is insufficient to offset the enthalpic driving force, and the solid phase remains favoured. At 273 K, the two contributions are exactly balanced ($\Delta G = 0$), marking the equilibrium melting point. Above this temperature, the entropic penalty outweighs the enthalpic driving force, rendering freezing non-spontaneous and the liquid phase thermodynamically favoured.

The Gibbs free energy equation provides a unified framework for evaluating reaction spontaneity by weighing the enthalpic and entropic contributions to the total free energy change. At elevated temperatures, the $T\Delta S$ term is amplified, increasing the thermodynamic reward for reactions that generate greater numbers of accessible microstates. This reflects the fact that at higher temperatures, the system possesses greater thermal energy to distribute across available microstates, making the free energy difference between high- and low-entropy states more pronounced. In this sense, the universe can be understood as evolving perpetually towards states of minimum free energy—that is, towards configurations with the greatest capacity for spontaneous change.

3.2 Free Energy & Equilibrium

Core Concept 3.5: Gibbs Free Energy & Equilibrium

We have established so far that if a reaction's ΔG is negative, it is favorable, and if it is positive, it is unfavorable. However, from EXTERNAL REFERENCE TO EQUILIBRIUM, we know that just because a reaction is favorable, it does not mean that it goes to completion. Many reactions find an equilibrium point where some reactants or products are still leftover although their presence is not favored in the context of ΔG .

There must thus be a quantitatively defined relationship between ΔG and K . Here, we introduce the **Van't Hoff Isotherm equation**.

$$\Delta G^\circ = -RT \ln K$$

Based on our previous establishment of $S = k_B \ln(W)$, we can see that entropy has a logarithmic relationship with the number of possible microstates (*which shift exponentially to the number molecules in a system*). Since K relates to the concentrations (*or the amounts*) of reactants and products, the accessible number of microstates change exponentially when Q shifts. Thus since ΔG incorporates ΔS into its value, the relationship between K and ΔG is logarithmic. Consequently, a linear change in ΔG leads to an exponential change in K .

Also recognize the relationship between ΔG and T . A change in T will have an effect on the magnitude of ΔG but also shift the value of K based on the ΔH of the reaction. The T at which K is exactly 1 would be when neither side of the reaction is inherently favorable and thus there an equal part reactant and product ratio.

Under non-standard conditions in the context of the current reactant product ratio Q . We would represent the total potential free energy left in the system by

$$\Delta G = \Delta G^\circ + RT \ln Q$$

or equivalently

$$\Delta G = -RT \ln \frac{K}{Q}$$

When $K = Q$, we have achieved equilibrium, thus there is no more potential free energy left to do work. If $K > Q$ or $K < Q$, the reaction still has free energy left to do work. In these cases the sign of ΔG represents the direction of the reaction that can perform work.

4 Examples and Applications

4.1 Work

Core Concept 4.1: Free Energy & Work

The free energy released from a system can be converted to **work** and used to **exert influence on an external body**. While reactions that are endothermic but entropy increasing enough to be favorable will be able to fulfill this purpose, it is more common to see reactions that are exothermic to be the source of work. In many cases, these exothermic reactions tend to decrease entropy along the process, thus the free energy that we can use to drive external processes will be less than the total heat released by the reaction.

This is what Gibbs free energy is used for in a practical sense. Let's take a look at the Gibbs free energy equation again.

$$\Delta G = \Delta H - T \Delta S$$

For $\Delta H < 0$, ΔH can be converted into work after "the entropy cost" is paid for.

For $\Delta S < 0$, ΔS must be compensated for in order to satisfy the second law of thermodynamics ("the entropy cost").

For $\Delta S > 0$, ΔS can also be converted into work.

Core Concept 4.2: Work Example (Rube Goldberg Machine)

Let's go through a collective example of how a system's change in entropy and enthalpy can be used to drive work onto an external system.

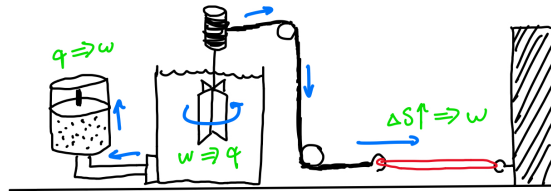


Figure 11: Entropy to Work Conversion

The figure above shows an initial state of a system. In this initial state, the red rubber band is in its stretched, low entropy position because there are fewer microstates in which the chains can be straight compared to its relaxed state where the chains are free to access a larger range of microstates because they aren't confined by the requirement to be straight.

ENTROPY TO WORK: As the rubber band contracts, the free energy provided by this rubber band system's entropy rise is converted into work that pulls the cable it is connected to.

WORK TO HEAT: As the work transferred from the rubber band enters the cable-paddle system (rotating the paddle), it immediately leaves the system as heat is inputted into the water system via friction from the paddle.

HEAT TO ENTROPY TO WORK: This heat energy is transferred onto a gas system within a piston chamber to push the piston up via the gas system's entropy increase due to an increase in volume.

*It is important to note that **the process of heat conversion to work is never a perfect conversion**. This is because the quality of energy from work is different from that of heat. Heat is the random vibration of molecules within a system, thus making it inherently high entropy. Work, on the other hand, is the organized motion of molecules within a system, thus making it low entropy. **If heat to work conversion is perfect, the universe will lose entropy in the process** (we know from the second law of thermodynamics that this is impossible), thus a portion of heat is released into the relatively lower temperature surrounding to compensate for the loss of entropy from the heat to work conversion process.*

Core Concept 4.3: Internal Energy

Internal energy (E) of a system is the sum of all kinetic and potential energies of all the system's components.

$$\Delta E = q + w$$

Energy gained or lost by the system is equal to the energy lost or gained by the surroundings.

To help us better understand this concept, imagine the transfer of heat energy into two gas systems to T_{final} from $T_{initial}$. We restrain one system to constant volume and the other to constant pressure.

CONSTANT VOLUME: The system is restricted from doing any expansion work to its surroundings, thus 100%

of the heat you add goes directly into increasing the E of the system (*raising T*).

CONSTANT PRESSURE: This system, on the other hand, is free to expand. Thus for the same T increase as the former system, you would need to add extra heat energy (*to raise T to T_{final}*) equivalent to the energy transferred to the surroundings from the expansion work it will do as volume increases.

It would take more heat energy to raise the temperature of the system with constant pressure than the system with constant volume by the same temperature increase.

4.2 Electrochemical Cells

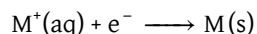
Core Concept 4.4: Electrochemical Cells

An electrochemical cell is a device that bridges the gap between chemical energy and electrical energy, effectively allowing us to use the free energy drop of a spontaneous chemical reaction to generate electrical energy for practical purposes. It works by physically separating a redox (reduction-oxidation; e^- transfer reactions) reaction so that e^- s are forced to travel through an external circuit rather than transferring directly between atoms within a solution.

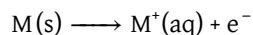
Core Concept 4.5: Electrode Potential

Before introducing electrochemical cells, we first need to understand electrode potentials. **Electrode potential** is a measure of the tendency of a metal to either "shed" its atoms into a solution as ions or "pull" ions from the solution to become part of the metal.

A high electrode potential means the metal is more likely to pull ions from the solution to become part of the metal. This is the reduction of metal ions into neutral metal atoms.



A low electrode potential means the metal is more likely to shed ions into a solution from the metal. This is the oxidation of neutral metal atoms into metal ions.



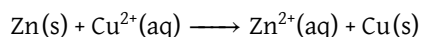
You cannot measure the potential of a single metal electrode in isolation. You always need at least one more metal electrode to compare it to. To solve this, chemists have designated a universal reference point to hydrogen called the **Standard Hydrogen Electrode (SHE)**, which is arbitrarily assigned a value of 0.00 V. Here are the electrode potentials of some common metals:

Potential	Reduction Half-Reaction
-3.05 V	$\text{Li}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Li}(\text{s})$
-2.71 V	$\text{Na}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Na}(\text{s})$
-2.37 V	$\text{Mg}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Mg}(\text{s})$
-1.66 V	$\text{Al}^{3+}(\text{aq}) + 3 \text{e}^- \longrightarrow \text{Al}(\text{s})$
-0.76 V	$\text{Zn}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Zn}(\text{s})$
-0.28 V	$\text{Ni}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Ni}(\text{s})$
0.00 V	$2 \text{H}^+(\text{aq}) + 2 \text{e}^- \longrightarrow \text{H}_2(\text{g})$
+0.34 V	$\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^- \longrightarrow \text{Cu}(\text{s})$
+0.80 V	$\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$
+1.06 V	$\text{Br}_2(\text{l}) + 2 \text{e}^- \longrightarrow 2 \text{Br}^-(\text{aq})$
+1.23 V	$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^- \longrightarrow 2 \text{H}_2\text{O}(\text{l})$
+1.36 V	$\text{Cl}_2(\text{g}) + 2 \text{e}^- \longrightarrow 2 \text{Cl}^-(\text{aq})$
+2.87 V	$\text{F}_2(\text{g}) + 2 \text{e}^- \longrightarrow 2 \text{F}^-(\text{aq})$

Core Concept 4.6: Galvanic Cell

We assign the units of V or J/C (joules per coulomb or energy per charge) because the potential tells us how much energy each e^- possesses as it spontaneously travels from a low potential electrode to a high potential electrode.

With that being said, if you were to place Zn metal powder into a solution of Cu ions, a spontaneous redox reaction would occur where the solid Zn metal would eventually turn into solid Cu.



Since this redox reaction is spontaneous ($\Delta G < 0$), free energy is released. A **galvanic cell** (a type of *electrochemical cell*) is a set up that allows us to capture this free energy and use it as electricity.

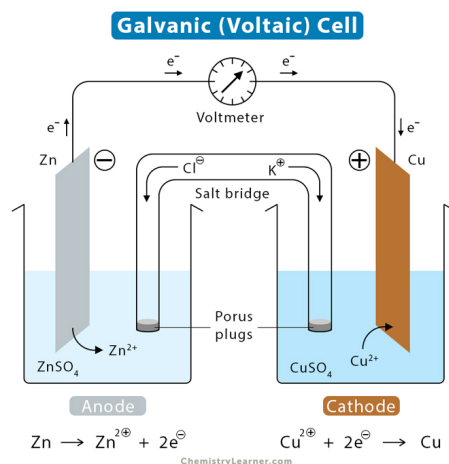


Figure 12: Galvanic Cell Diagram

In a galvanic cell set up, we call the low and high electrode potential metal the anode and cathode, respectively. In the above schematic, the anode is Zn and the cathode is Cu. Since the solutions of the two electrodes are physically separated, the presence of a salt bridge allows each solution to maintain electrical neutrality and keep the redox reaction moving forward rather than slow down and eventually stop due to the eventual build up of one-sided charge (positive ions on the anode from $\text{Zn}^{2+}(\text{aq})$ accumulation and negative ions on the cathode from SO_4^{2-} presence with the decreasing amounts of $\text{Cu}^{2+}(\text{aq})$). The result is a system that captures the potential difference between the two electrodes until the redox reaction is complete, at which point the anode is completely in its aqueous ion form and the cathode is completely in its solid form.

We can express the free energy gained from this system via the following equation:

$$\Delta G^\circ = -nFE^\circ$$

where n is the number of moles of e^- , F is the Faraday constant which measures the amount of charge per mole of e^- (96485 C mol^{-1}), and E° is cell potential or the difference in potential between the anode and the cathode (in the above case it would be $0.34\text{V} - (-0.80\text{V}) = 1.14\text{V}$).

Let's go through a calculation for the galvanic cell above ($\text{Zn}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \longrightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu}(\text{s})$).

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ = 0.34\text{V} - (-0.80\text{V}) = 1.14\text{V}$$

$n = 2\text{mol}$ because two e^- s are being transferred across the cell. Thus

$$\Delta G^\circ = -(2\text{ mol})(96485 \text{ C mol}^{-1})(1.14 \text{ J J}^{-1}) = -219.98 \text{ kJ}$$

It's important to keep in mind that this free energy value is for every mole of the redox reaction that occurs, thus the same galvanic cell with Zn as its limiting reactant would have a free energy value of -219.98 kJ per mole of Zn.

Core Concept 4.7: Concentration Cells

Now we introduce a **concentration cell**, an electrochemical cell that uses the same metal for both the anode and cathode but at different solution concentrations.

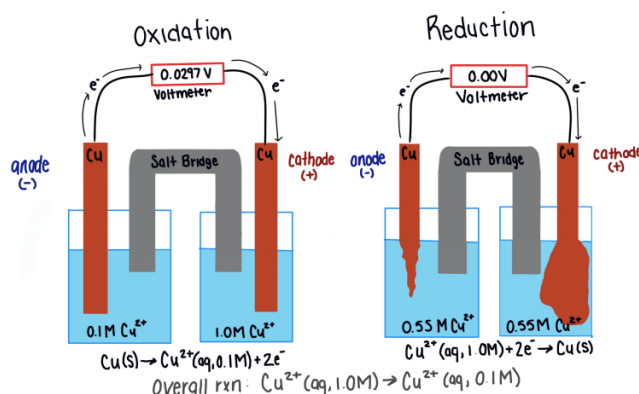
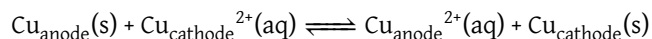


Figure 13: Concentration Cell Diagram

Let's take a look at the chemical equation for the redox reaction in the concentration cell.



The Eq for this reaction is $K = 1$, however

$$Q = \frac{[\text{Cu}_{\text{anode}}^{2+}]}{[\text{Cu}_{\text{cathode}}^{2+}]} < 1$$

Thus, the forward reaction is favored, raising the concentration of the anode solution at the expense of that of the cathode. Via the concentration differences, e^- 's flow just like it did in the galvanic cell and we are able to capture the free energy of the redox reaction.

Quantitatively, we can measure the cell potential in volts using the following algebraic manipulations:

$$(STANDARD\ CONDITIONS) \quad \Delta G^\circ = -nFE^\circ = -RT\ln(K)$$

$$(NONSTANDARD\ CONDITIONS) \quad \Delta G = -nFE$$

$$\Delta G = \Delta G^\circ - (-RT\ln(Q))$$

$$-nFE = -nFE^\circ + RT\ln(Q)$$

$$E = E^\circ - \frac{RT}{nF} \ln(Q)$$

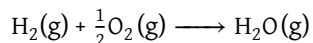
Since E° is always 0 for concentration cells and $\ln(x) = 2.303 \times \log_{10}(x)$, we can simplify the above equation to

$$E = -\frac{0.0592}{n} \log\left(\frac{[M_{\text{anode}}^{n+}]}{[M_{\text{cathode}}^{n+}]}\right)$$

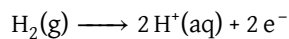
In our example above, $E = 0.0296V$ for $[\text{Cu}_{\text{anode}}^{2+}] = 0.1M$ and $[\text{Cu}_{\text{cathode}}^{2+}] = 1.0M$ (or $Q = 0.1$). Because the potential is low compared to that of a galvanic cell, it does not actually provide any practical value in the context of batteries.

Core Concept 4.8: Hydrogen Fuel Cell

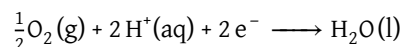
Okay, now we move onto one last type of electrochemical cell called hydrogen fuel cells. **Hydrogen fuel cells** does not use the typical metal elements as the anode and cathode, but rather uses the **hydrogen gas** as the anode and **oxygen gas** as the cathode. It simply utilizes the e^- transfer energy during the process of forming water from these gases.



When we break this chemical reaction down into components we get the anode reaction (oxidation)



and the cathode reaction (reduction)



Here is a schematic of a hydrogen fuel cell.

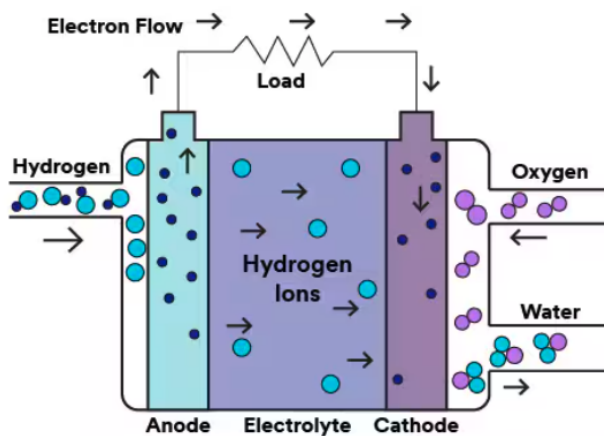


Figure 14: Hydrogen Fuel Cell Diagram

We can quantitatively express the free energy of the hydrogen fuel cell using the same method as we did for the galvanic cell.

$$E_{\text{cell}}^{\circ} = E_{\text{oxygen}}^{\circ} - E_{\text{hydrogen}}^{\circ} = 1.23\text{V} - (-0.00\text{V}) = 1.23\text{V}$$

$$\Delta G^{\circ} = -(2\text{ mol})(96485\text{ C mol}^{-1})(1.23\text{ J J}^{-1}) = -237.35\text{ kJ}$$

Thus the free energy of a hydrogen fuel cell per mole of H_2 is -237.35 kJ .